University of BRISTOL

Lecture 2
Elastically suspended propeller
Part 1 Modelling

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This lecture

- Motivation
  - Applications
- Our modelling building blocks
  - Angular momentum and gyroscopic moment
  - Other disciplines (aerodynamics, structures)
- Reed's model
  - EoM derivation (FBD)
  - Reed-Ribner's aerodynamics
- Equations of motion
  - Component matrices and model assembly
  - Annotated model in Matlab
  - Initial ODE solution (whirling response)

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Motivation

Lockheed L-188 Electra case: *Beyond the Horizons - The Lockheed Story* by Walter J. Boyne (St. Martin's Press, New York 1998)

commons.wikimedia.org/wiki/File:Lockheed_L-188_Electra_Eastern_Air_Lines.jpg

"… In a normal aircraft, the whirl mode could operate only within the limits of the flexibility of the engine mounts. If, however, some structural element of the power plant, the power-plant mounting system, or the nacelle was in a damaged or weakened condition, the whirl mode would not damp out, but could become more violent, increasing damage to the structure, and could approach the natural frequency of the wing. This would perpetuate the whirl mode in a form of induced flutter and lead to catastrophic failure.

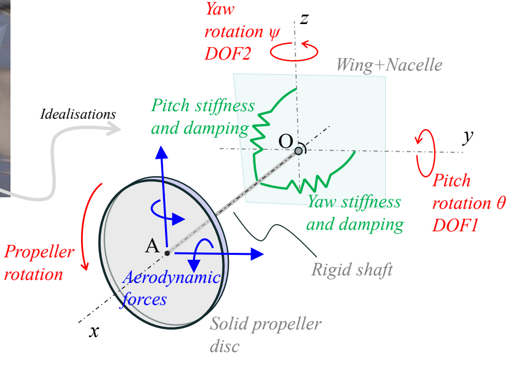
*John Margwarth, another University of Michigan man, was director of safety for Lockheed, and it was his insight that led to an investigation revealing that the Electra's fatal flaw was in the three member structure connecting the gearbox and the engine, a part supplied by the engine manufacturers. When **one member of that structure failed**, the engine mount became flexible. On an outboard engine, at the Electra's original cruise speed, failure of the strut induced immediate, violent flutter that tore the wing off. …"*

Modern applications: fix wing propellers

Problem modelling philosophy

The objective is to develop the smallest model (minimal model) that can present the characteristics of whirling wing-nacelle-propeller previously observed in industrial propeller installations (*see* L-188 Electra case). Such model was proposed by Houbolt and Reed* in the form of a 2DOF model shown below.

Reed's model **of elastically suspended rigid propeller**



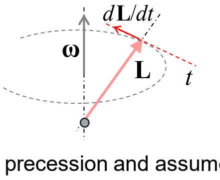
*W.H. Reed III, S.R. Bland, An analytical treatment of aircraft propeller precession instability, NASA TN D-659, 1961
J.C. Houbolt, W.H. Reed III, Propeller-nacelle whirl flutter, Journal of the Aerospace Sciences, Vol.29, No.3, 1962

Gyroscopic effect

Similar to the non-inertial adaption used to introduce the centrifugal and Coriolis effects in case of rotating elastic blades earlier, here we introduce the concept of *gyroscopic moment* M_{gyro}. It concerns the case of a *precessing* rigid propeller (*i.e.* motion of the rotating propeller about another axis)

Moment **M** causing precession (a system of external actions causing disturbance to **L**, assume **L** in the body frame):

$$\mathbf{M} = \frac{d\mathbf{L}}{dt}$$



Consider above scenario involving precession and assume no or negligible change in magnitude of **L**:

$$\frac{d\mathbf{L}}{dt} = \dot{\mathbf{L}}\big|_{body} + \boldsymbol{\omega}\times\mathbf{L} \approx \boldsymbol{\omega}\times\mathbf{L}$$

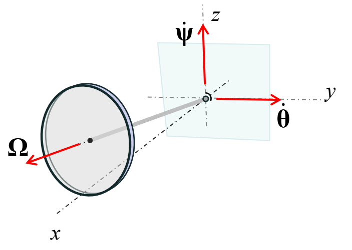
$$\mathbf{M} - \boldsymbol{\omega}\times\mathbf{L} = \mathbf{0} \rightarrow \mathbf{M} + \mathbf{M}_{gyro} = 0$$

Looking for an inertial effect to present as a moment (or force) arising from the presence of precessional motion.

Gyroscopic moment $$\mathbf{M}_{gyro} = -\boldsymbol{\omega}\times\mathbf{L} = \mathbf{L}\times\boldsymbol{\omega}$$

Gyroscopic effect

Introduce the *gyroscopic moment* to the Reed's propeller configuration. Further, *assume small vibrations* – all precessional angular speed components aligned with the original frame of reference:



Angular momentum **L** (*neglect pitch/yaw **L** contribution*) and instantaneous angular speed **ω**:

$$\mathbf{L} = \mathbf{L}_{\Omega} + \mathbf{L}_{\theta} + \mathbf{L}_{\psi} \approx \mathbf{L}_{\Omega},\ \mathbf{L}_{\Omega} = I_x \boldsymbol{\Omega}$$

$$\boldsymbol{\omega} \approx \boldsymbol{\Omega} + \dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\psi}}$$

Gyro moment in Reed's configuration:

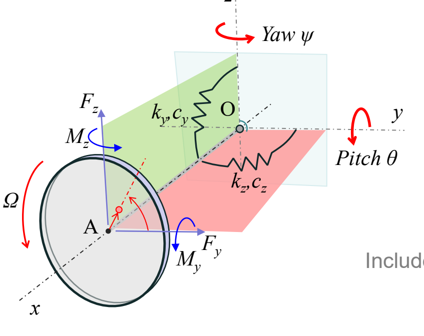
$$\mathbf{M}_{gyro} = \mathbf{L} \times \boldsymbol{\omega} = I_x \boldsymbol{\Omega} \times (\boldsymbol{\Omega} + \dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\psi}}) = I_x \boldsymbol{\Omega} \times \dot{\boldsymbol{\theta}} + I_x \boldsymbol{\Omega} \times \dot{\boldsymbol{\psi}} = \mathbf{M}_{gyro,\theta} + \mathbf{M}_{gyro,\psi}$$

$$\mathbf{M}_{gyro} = \mathbf{M}_{gyro,\theta} + \mathbf{M}_{gyro,\psi}$$

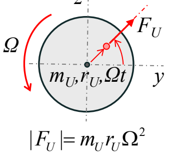
Include this in Reed's model's FBD when deriving its EoMs.

Adopted Reed's model configuration

All quantities established here follow the "right-hand rule". Note, this is different from the classical Reed's model where some conventions and sign differences (θ angle) with the present approach occur. Notation, frame of reference and sign conventions adopted here are as follows:



Include unbalance excitation:



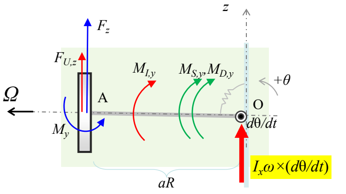
$$|F_U| = m_U r_U \Omega^2$$

Free body diagrams

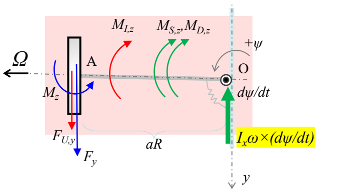
Derivation of the EoMs using free body diagrams in the two perpendicular planes (X-Z and X-Y).

Equilibrium in X-Z plane:

Note the colour code aligned with previous slide.



Equilibrium in X-Y plane:



Sum of all moments about O:

$$M_{I,y} + M_{S,y} + M_{D,y} + M_{gyro,\theta} - M_y + (F_z + F_{U,z})aR = 0$$

$$M_{I,z} + M_{S,z} + M_{D,z} - M_{gyro,\psi} - M_z - (F_y + F_{U,y})aR = 0$$

where

$$M_I = I_{ref}\ddot{\alpha}, M_D = c_\alpha \dot{\alpha}, M_S = k_\alpha \alpha, F_U = m_U r_U \Omega^2$$

Equations of motion:

$$I_O\ddot{\theta} + c_\theta\dot{\theta} + I_x\Omega\dot{\psi} + k_\theta\theta = M_{A,\theta} - aR\,F_U\sin(\Omega t) \qquad M_{A,\theta} = M_y - aR\,F_z$$

$$I_O\ddot{\psi} + c_\psi\dot{\psi} - I_x\Omega\dot{\theta} + k_\psi\psi = M_{A,\psi} + aR\,F_U\cos(\Omega t) \qquad M_{A,\psi} = M_z + aR\,F_y$$

Introduction to aerodynamic loads

The model adopted in this section has two key attributes. Firstly, it is **linear** (perturbed model from the equilibrium condition). Such models are suitable for (linear) stability analysis and not suitable for the structural performance analysis. Secondly, this model draws from the **quasi-steady aerodynamics** blade segment theories.

To illustrate the process, consider the Taylor expansion of a quantity Q=f(a,b):

$$Q(a,b) = Q(a_0,b_0) + \left.\frac{\partial Q}{\partial a}\right|_{a_0,b_0}(a-a_0) + \left.\frac{\partial Q}{\partial b}\right|_{a_0,b_0}(b-b_0) + h.o.t.$$

$$\Delta Q \approx \left.\frac{\partial Q}{\partial a}\right|_{a_0,b_0}\Delta a + \left.\frac{\partial Q}{\partial b}\right|_{a_0,b_0}\Delta b \;= Q_a\Delta a + Q_b\Delta b \;= \begin{bmatrix} Q_a & Q_b \end{bmatrix}\begin{bmatrix}\Delta a \\ \Delta b\end{bmatrix}$$

Aerodynamic derivatives: $responses \in \{F_y, F_z, M_y, M_z\}, \quad states \in \{\theta, \psi, \dot{\theta}, \dot{\psi}\}$

Aerodynamic derivatives	Pitch (θ)	Yaw (ψ)	d(Pitch)/dt (q)	d(Yaw)/dt (r)
Fy (y)	c yθ	c yψ	c yq	c yr
Fz (z)	c zθ	c zψ	c zq	c zr
My (m)	c mθ	c mψ	c mq	c mr
Mz (n)	c nθ	c nψ	c nq	c nr

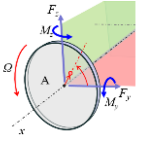
Table pruning: 16 derivatives overall; reduced to 8 due to symmetries (see colours); **reduced to 6 due to smallness** (see grey tones). Diagonal symmetry plus; antidiagonal symmetry minus.

Reed-Ribner aerodynamic model

A frequently applied model that belongs to the category of the *quasi-steady linear aerodynamic models* is the model originally developed by Reed while following Ribner's approach to the air flow around a "propeller in yaw" *.

J.C. Houbolt, W.H. Reed III, Propeller-nacelle whirl flutter, *Journal of the Aerospace Sciences*, Vol.29, No.3, 1962

Reed's model (adjusted to comply with our θ sign convention):



$$F_y = (N_b/2)K_a(A_1'\psi - aA_1(\dot{\psi}/\Omega) - A_2(\dot{\theta}/\Omega))$$
$$F_z = (N_b/2)K_a(-A_1'\theta + aA_1(\dot{\theta}/\Omega) - A_2(\dot{\psi}/\Omega))$$
$$M_y = (N_b/2)K_a R(A_2'\psi - aA_2(\dot{\psi}/\Omega) - A_3(\dot{\theta}/\Omega))$$
$$M_z = (N_b/2)K_a R(-A_2'\theta + aA_2(\dot{\theta}/\Omega) - A_3(\dot{\psi}/\Omega))$$

A.D.	(θ)	(ψ)	(q)	(r)
Fy (y)	c yθ	c yψ	c yq	c yr
Fz (z)	c zθ	c zψ	c zq	c zr
My (m)	c mθ	c mψ	c mq	c mr
Mz (n)	c nθ	c nψ	c nq	c nr

Aerodynamic integrals:

$$A_1 = \frac{c}{R}\int_0^1 \frac{\mu^2}{\sqrt{\mu^2+\eta^2}}d\eta,\ A_1' = \mu A_1,\ A_2' = \frac{c}{R}\int_0^1 \frac{\mu^2\eta^2}{\sqrt{\mu^2+\eta^2}}d\eta,\ A_3 = \frac{c}{R}\int_0^1 \frac{\eta^4}{\sqrt{\mu^2+\eta^2}}d\eta$$

Constants:

$$K_a = \frac{1}{2}\rho c_{l,\alpha} R^4\Omega^2,\ \mu = \frac{V}{\Omega R}$$

* *H.S. Ribner, Propeller in yaw, NACA Rep. 820, 1945*

Further development of aerodynamic RHS

As indicated by the model developed above (slide 9), all individual aerodynamic load components need to be combined to produce the overall pitch and yaw moment loading.

Overall aerodynamic moments:

$$M_{A,\theta} = M_y - aR\,F_z$$
$$M_{A,\psi} = M_z + aR\,F_y$$

Substituting (θ-adjusted) Reed-Ribner aerodynamic loads:

$$M_\theta = \frac{N_b}{2}K_a R\left[-(A_3 + a^2A_1)\frac{\dot{\theta}}{\Omega} + A_2'\psi + aA_1'\theta\right]$$
$$M_\psi = \frac{N_b}{2}K_a R\left[-(A_3 + a^2A_1)\frac{\dot{\psi}}{\Omega} - A_2'\theta + aA_1'\psi\right]$$

Transformation to (linear) vector-matrix form to identify the aerodynamic matrices:

$$\begin{bmatrix} M_\theta \\ M_\psi \end{bmatrix} = \frac{N_b}{2}K_a R\begin{bmatrix} aA_1' & A_2' \\ -A_2' & aA_1' \end{bmatrix}\begin{bmatrix} \theta \\ \psi \end{bmatrix} + \frac{N_b}{2}K_a\frac{R}{\Omega}\begin{bmatrix} -(A_3+a^2A_1) & 0 \\ 0 & -(A_3+a^2A_1) \end{bmatrix}\begin{bmatrix} \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

$$\mathbf{M}_{aer} = \begin{bmatrix} M_\theta \\ M_\psi \end{bmatrix} = -\mathbf{K}_{aer}\begin{bmatrix} \theta \\ \psi \end{bmatrix} - \mathbf{C}_{aer}\begin{bmatrix} \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

Assembling full Reed's model

Consider previously derived EoMs:

$$\begin{bmatrix} I_{yy} & 0 \\ 0 & I_{zz} \end{bmatrix}\begin{bmatrix} \ddot{\theta} \\ \ddot{\psi} \end{bmatrix} + \begin{bmatrix} c_\theta & I_x\Omega \\ -I_x\Omega & c_\psi \end{bmatrix}\begin{bmatrix} \dot{\theta} \\ \dot{\psi} \end{bmatrix} + \begin{bmatrix} k_\theta & 0 \\ 0 & k_\psi \end{bmatrix}\begin{bmatrix} \theta \\ \psi \end{bmatrix} = \begin{bmatrix} M_y - aR\,F_z \\ M_z + aR\,F_y \end{bmatrix} + aR\,F_U \begin{bmatrix} -\sin(\Omega t) \\ \cos(\Omega t) \end{bmatrix}$$

Substitute Reed-Ribner aerodynamic loads (and use smaller font due to sheer size):

$$\begin{bmatrix} I_{yy} & 0 \\ 0 & I_{zz} \end{bmatrix}\begin{bmatrix} \ddot{\theta} \\ \ddot{\psi} \end{bmatrix} + \begin{bmatrix} c_\theta & I_x\Omega \\ -I_x\Omega & c_\psi \end{bmatrix}\begin{bmatrix} \dot{\theta} \\ \dot{\psi} \end{bmatrix} + \begin{bmatrix} k_\theta & 0 \\ 0 & k_\psi \end{bmatrix}\begin{bmatrix} \theta \\ \psi \end{bmatrix} = \frac{N_b}{2}K_a R \begin{bmatrix} aA_1' & A_2' \\ -A_2' & aA_1' \end{bmatrix}\begin{bmatrix} \theta \\ \psi \end{bmatrix} + \frac{N_b}{2}K_a \frac{R}{\Omega}\begin{bmatrix} -(A_3+a^2A_1) & 0 \\ 0 & -(A_3+a^2A_1) \end{bmatrix}\begin{bmatrix} \dot{\theta} \\ \dot{\psi} \end{bmatrix} + aR\,F_U \begin{bmatrix} -\sin(\Omega t) \\ \cos(\Omega t) \end{bmatrix}$$

Rearrange equations and collect similar terms:

$$\begin{bmatrix} I_{yy} & 0 \\ 0 & I_{zz} \end{bmatrix}\begin{bmatrix} \ddot{\theta} \\ \ddot{\psi} \end{bmatrix} + \left(\begin{bmatrix} c_\theta & 0 \\ 0 & c_\psi \end{bmatrix} + \begin{bmatrix} 0 & I_x\Omega \\ -I_x\Omega & 0 \end{bmatrix} + \frac{N_b}{2}K_a \frac{R}{\Omega}\begin{bmatrix} A_3+a^2A_1 & 0 \\ 0 & A_3+a^2A_1 \end{bmatrix}\right)\begin{bmatrix} \dot{\theta} \\ \dot{\psi} \end{bmatrix} + \left(\begin{bmatrix} k_\theta & 0 \\ 0 & k_\psi \end{bmatrix} + \frac{N_b}{2}K_a R \begin{bmatrix} -aA_1' & -A_2' \\ A_2' & -aA_1' \end{bmatrix}\right)\begin{bmatrix} \theta \\ \psi \end{bmatrix} = aR\,m_U r_U \Omega^2 \begin{bmatrix} -\sin(\Omega t) \\ \cos(\Omega t) \end{bmatrix}$$

Inertia | structural damping | gyro effect | aerodynamic damping | structural stiffness | aerodynamic stiffness | unbalance loading

Full assembled Reed's model with unbalance loading:

$$\mathbf{M}\begin{bmatrix} \ddot{\theta} \\ \ddot{\psi} \end{bmatrix} + \left(\mathbf{C}_S + \mathbf{C}_{gyro} + \mathbf{C}_{aer}\right)\begin{bmatrix} \dot{\theta} \\ \dot{\psi} \end{bmatrix} + \left(\mathbf{K}_S + \mathbf{K}_{aer}\right)\begin{bmatrix} \theta \\ \psi \end{bmatrix} = \mathbf{f}(t;\Omega)$$

Compact and standard form of Reed's model:

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{C}\dot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{f}(t;\Omega), \quad \mathbf{q}(t) = [\theta(t), \psi(t)] = ?$$

Annotated Reed's model in Matlab

Copy and paste to Matlab editor, save as "batch_reed.m" and then execute to study the whirl dynamics.

Problem parameters

```
function batch_reed
% Parameters
R=0.152;          % m, Rotor radius
Omega=40;         % rad*s^(-1}, Rotor angular velocity
a=0.25;           % - ,Pivot length to rotor radius ratio
Ix=0.000103;      % kg*m^2, Rotor moment of inertia
In=0.000178;      % kg*m^2, Nacelle moment of inertia
Ctht=0.001;       % N*m*s*rad^{-1}, Structural pitch damping
Ktht=0.4;         % N*m*rad^{-1}, Structural pitch stiffness
Cpsi=0.001;       % N*m*s*rad^{-1}, Structural yaw damping
Kpsi=0.4;         % N*m*rad^{-1}, Structural yaw stiffness
Nb=4;             % -, Number of blades
c=0.026;          % m, Blade chord
cla=2*pi;         % rad^{-1}, Blade lift slope
rho=1.225;        % kg*m^{-3}, Air density SLS
m_u=0.001;        % kg, unbalance mass
r_u=0.001;        % m, unbalance radius/offset
V=1.37*Omega*R; % m*s^{-1}, Freestream velocity (nominal V=6.7!)
```

Explore the influence of the support system

Explore the influence of the varying velocity

Aerodynamic integrals

```
% Aerodynamic constants and integrals
Ka1=0.5*rho*cla*R^4*Omega^2; COEF=(Nb/2)*Ka1*R; mu=V/(Omega*R);
A1=(c/R)*quad(@(eta1)(mu.^2./sqrt(mu.^2+eta1.^2)),0,1); Ap1=mu*A1;
Ap2=(c/R)*quad(@(eta1)(mu.^2.*eta1.^2./sqrt(mu.^2+eta1.^2)),0,1);
A3=(c/R)*quad(@(eta1)(eta1.^4./sqrt(mu.^2+eta1.^2)),0,1);
```

Model matrices

```
% Matrices
Ms=diag([In,In]); % Mass
Cs=diag([Ctht,Cpsi]); % Structural damping
Cg=Omega*[0,Ix;-Ix,0];% Gyroscopic
Ks=diag([Ktht,Kpsi]); % Stiffness
Kaer=COEF*[-a*Ap1,-Ap2;Ap2,-a*Ap1]; % Aerodynamic stiffness
Caer=(COEF/Omega)*[A3+a^2*A1,0;0,A3+a^2*A1]; % Aerodynamic damping
```

Note the constituent physical sources

Model assembly

```
% Assembled Reed's model system and state space matrix
MM=Ms; CC=Cs+Cg+Caer; KK=Ks+Kaer;
AA=[zeros(2),eye(2);-inv(MM)*KK,-inv(MM)*CC];
```

ODE solver

```
% ODE solver (time, initial conditions, unbalance forcing)
[TT,XX]=ode45(@l_model,[0,5],[0;0.01;1;0]);
```

Explore the influence of different initial conditions

Postprocessing and/or plotting

```
% Basic plotting
figure
subplot(2,1,1),plot(TT,XX(:,1:2))
subplot(2,1,2),plot(TT,XX(:,3:4))
figure
subplot(2,2,1),plot(TT,-a*R*XX(:,1)), ylabel('z_A'), grid
subplot(2,2,4),plot(a*R*XX(:,2),TT), xlabel('y_A'), grid
subplot(2,2,2),title('Orbit A'), comet(a*R*XX(:,2),-a*R*XX(:,1)), grid
```

Observe orbit, phase, frequency, amplitude, decay, …

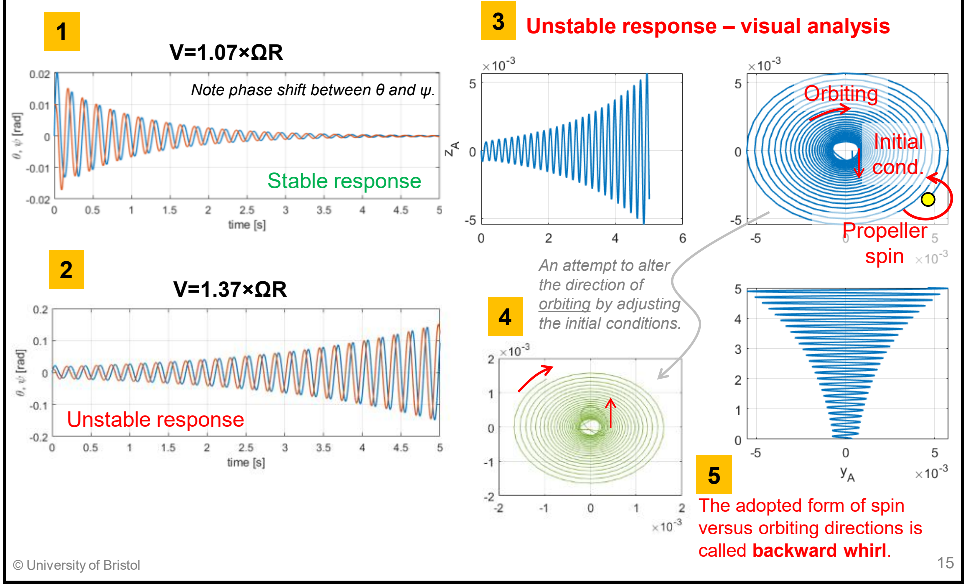
Model subfunction

```
    function dxx=l_model(tt,xx)
        % State space model for ODE solver
        ff=(m_u*r_u*Omega^2)*[-sin(Omega*tt);cos(Omega*tt)]; % unbalance
        gg=[[0;0];inv(MM)*ff]; % state space model RHS
        dxx=AA*xx+gg; % state space equation
    end
end
```

Note model in the 1st order ODE form (see slide 16)

Initial numerical exploration

Zero unbalance forcing and nonzero initial conditions – two distinct behaviours.



Summary

- Appreciate notable features of the model and their functional purpose linked to the applications.
- Remember the origins and specifics of the modelled aerodynamic and gyroscopic effects – Reed-Ribner model and gyroscopic moments.
- Understand the FBD-based solution process leading to the Reed's model and physical interpretation of the component matrices.
- Study closely time domain responses and observe stability, phase shift, whirling behaviour, unbalance forcing, impact of initial conditions, etc.

Appendix: Transformation to the 1st order form

To solve 2nd order ODEs numerically, we tend to express the equations in the 1st order ODE form. Transformation from the *physical space* to the *state space* realises this operation. 1st order ODE systems are solvable using generic numerical integration (time stepping) solvers in various computational environments such as Matlab, Python, etc.

General EOM rearranged:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{f}(t)$$ *2nd order ODEs*

$$\mathbf{M}\ddot{\mathbf{q}} = -\mathbf{C}\dot{\mathbf{q}} - \mathbf{K}\mathbf{q} + \mathbf{f}(t)$$

$$\ddot{\mathbf{q}} = -\mathbf{M}^{-1}\mathbf{C}\dot{\mathbf{q}} - \mathbf{M}^{-1}\mathbf{K}\mathbf{q} + \mathbf{M}^{-1}\mathbf{f}(t)$$

$$\ddot{\mathbf{q}} = \begin{bmatrix} -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ \dot{\mathbf{q}} \end{bmatrix} + \mathbf{M}^{-1}\mathbf{f}(t)$$

Mathematical identity to augment the EOM:

$$\dot{\mathbf{q}} = \dot{\mathbf{q}}$$

$$\dot{\mathbf{q}} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ \dot{\mathbf{q}} \end{bmatrix} + \mathbf{0}$$

Combine the two systems of equations:

$$\begin{bmatrix} \dot{\mathbf{q}} \\ \ddot{\mathbf{q}} \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ \dot{\mathbf{q}} \end{bmatrix} + \begin{bmatrix} \mathbf{0} \\ \mathbf{M}^{-1}\mathbf{f}(t) \end{bmatrix}$$

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{g}(t)$$ *1st order ODEs*

$$\mathbf{x} = \begin{bmatrix} \mathbf{q} \\ \dot{\mathbf{q}} \end{bmatrix}, \mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{C} \end{bmatrix}, \mathbf{g}(t) = \begin{bmatrix} \mathbf{0} \\ \mathbf{M}^{-1}\mathbf{f}(t) \end{bmatrix}$$